Digital Twin of a Distribution Transformer for Condition Monitoring and Predictive Maintenance

1. Challenge Overview

Distribution transformers are critical components of the electrical power distribution network. They operate continuously under varying load conditions, temperature changes, voltage fluctuations, and environmental stresses.

Traditional maintenance is often performed at fixed intervals, regardless of the actual condition of the transformer. This can lead to:

- Unexpected transformer failures
- Increased maintenance costs
- Power interruptions
- Reduced asset life

A **Digital Twin** is a virtual representation of a physical asset that continuously uses operational data to simulate, monitor, and predict its behavior. By creating a Digital Twin of a distribution transformer, utilities can monitor asset health, detect faults early, estimate remaining useful life, and plan maintenance more effectively.

2. Problem Statement

Develop a Digital Twin prototype of a distribution transformer that can:

- Simulate transformer behavior using operational data.
- Monitor key electrical and thermal parameters.
- Estimate transformer health conditions.
- Detect abnormal operating conditions.
- Provide predictive maintenance recommendations.
- Display results through an interactive dashboard.

The solution should work with simulated datasets provided during the hackathon and should be designed so that it can be extended later to connect with real sensors and utility monitoring systems.

3. Key Objectives

Your solution should address the following objectives:

Health Monitoring

Monitor important transformer parameters such as:

- Voltage
- Current
- Loading
- Oil temperature
- Ambient temperature

Health Assessment

Develop a method to estimate a **Transformer Health Index** that indicates the overall condition of the asset.

Anomaly Detection

Identify abnormal operating conditions and generate alerts before failures occur.

Predictive Maintenance

Provide recommendations on when maintenance actions may be required.

Visualization

Create a dashboard that presents transformer condition, performance trends, alerts, and maintenance recommendations in an easy-to-understand format.

4. Expected Solution Architecture

A typical solution may include the following modules:

Data Input Layer

- Simulated sensor data
- Historical datasets
- CSV files
- IoT data simulator

Digital Twin Engine

- Mathematical model of transformer behavior
- AI/ML models for prediction
- Condition assessment logic

Analytics Layer

- Health Index calculation
- Fault prediction
- Anomaly detection
- Remaining useful life estimation

Dashboard Layer

- Real-time monitoring
- Health score visualization
- Alerts and notifications
- Maintenance recommendations

Reporting Layer

- Trend analysis
- Performance reports
- Downloadable summaries

5. Inputs Available

Participants may use:

Transformer Data

- Voltage
- Current
- Loading information
- Oil temperature
- Ambient temperature

Asset Information

- Transformer ratings
- Nameplate specifications

Operational Data

- Historical records
- Simulated SCADA-style datasets
- Open-source datasets

6. Expected Deliverables

Teams should submit:

Software Prototype

A working Digital Twin application demonstrating transformer monitoring and analysis.

Dashboard

An interactive dashboard showing:

- Transformer loading
- Temperatures
- Health Index
- Alerts
- Maintenance recommendations

Technical Documentation

- Solution architecture
- Methodology
- Assumptions
- User guide

Source Code

Complete project source code with instructions for execution.

Demonstration

- Short presentation
- Live demo or recorded walkthrough